Python Programming - 2301CS404

Lab - 7

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01) WAP to find sum of tuple elements.

Sample Input:

```
T = (10, 20, 30, 40)
```

Sample Output:

```
Sum of tuple elements = 100
```

In [6]:
```python
# t1 = tuple( int(i) for i in input("Enter Elements:").split(" "))
# t1

# or

t1 = tuple( map( int , input("Enter Elements").split()))
t1
print("Sum of tuple elements :",sum(t1))
```

```
Sum of tuple elements : 100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

```
T = (7, 2, 9, 4, 1, 5)
K = 2
```

Sample Output:

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

In [7]:
```python
t1=tuple(map(int,input("Enter a list:").split()))
k=int(input("Enter k"))
print(f"Minimum {k} elements: {tuple(sorted(t1))[:k]}")
print(f"Maximum {k} elements: {tuple(sorted(t1))[-k:][::-1]}")
```

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

In [16]:
```python
length = int(input("Enter length of List:"))
l1=[]
l2=[]
for i in range(0 , length):
    t1=tuple(map(int,input("Enter a list:").split()))
    l1.append(t1)
    if ( all(x%k==0 for x in t1)):
        l2.append(t1)
print(l1)
print("Tuples divisible by 2 :",l2)
```

```
[(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 12)]
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

In [20]:
```python
# l1=list(map(int,input("Enter a list:").split()))
# result=[]
# for i in l1:
#     t1 = (i , i**3)
```

```
#     result.append(t1)
# print("List of tuples with number and its cube:" , result)

l1=list(map(int,input("Enter a list:").split()))
print("List of tuples with number and its cube:" ,list((i ,i**3) for i in l1))
```

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

In [24]:

```
length = int(input("Enter length of List:"))
l1=[]
l2=[]
for i in range(0 , length):
    t1=tuple(map(int,input(f"Enter a list {i}:").split()))
    l1.append(t1)
    if ( all( x > 0 for x in t1)):
        l2.append(t1)
print(l1)
print("Tuples with all positive elements:",l2)
```

```
[(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

```
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]
```

In [28]:
```
length = int(input("Enter length of List:"))
l1=[]
for i in range(0 , length):
    t1=tuple(map(int,input(f"Enter a tuple of list {i}:").split()))
    l1.append(t1)

print("original tuple list :" , l1)

for i in l1:
    if ( len(i) == 2):
        l1.remove(i)
print("List after removing tuples of length 2 :" ,l1 )
```

```
original tuple list : [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8, 9)]
```

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

In [32]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
x=int(input("Enter x :"))
```

```
t2= list(t1)
for i in t2:
    if i == x:
        t2.remove(i)
t1 = tuple(t2)
# # print(t1)
# t2 = tuple()
# for i in t1:
#     if i != x:
#         t2 += (i,)
# print(t2)
```

```
(5, 7, 9)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

In [33]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
t1= set(t1)
print(t1)
```

```
{9, 3, 5, 7}
```

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

In [35]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
t2=tuple()
for i in range(0 , len(t1)-1):
    t2 += (t1[i]*t1[i+1])
print("Resultant tuple after multiplying adjacent elements:",)
```

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

In [36]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
t2=tuple(map(int,input(f"Enter a tuple:").split()))
if len(t1) == len(set(t1)):
    print(t1,"is distinct? -> True")
else:
    print(t1,"is distinct? -> False")

if len(t2) == len(set(t2)):
    print(t2,"is distinct? -> True")
else:
    print(t2,"is distinct? -> False")
```

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 3, 2) is distinct? -> False
```

11) WAP to rotate the elements of a tuple to the right by `k` positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

In [43]:
```
t1=tuple(map(int,input("Enter a tuple:").split()))
k = int(input("Enter value of K :"))
t2 = (t1[-k:] + t1[:k+1])
print(t2)
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

In [46]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
t2=tuple(map(int,input(f"Enter a tuple:").split()))
result=[]
if len(t1) != len(t2):
    print("Length not Equal")
for i in range(0 , len(t1)):
```

```
    result.append(t1[i])
    result.append(t2[i])

result = tuple(result)
print(result)
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

In [49]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
result = tuple( i**2 for i in t1 if i>10 and i%2==0)
print(result)
```

```
(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

In [51]:
```
t1=tuple(map(int,input(f"Enter a tuple:").split()))
result = []
```

```
for i in range(len(t1)):
    num = t1[i]
    if num > 1:
        is_prime = True
        for j in range(2, num):
            if num % j == 0:
                is_prime = False
                break
        if is_prime:
            result.append((i, num))

print(tuple(result))
```

```
((1, 7), (3, 11))
```

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

In [4]:
```
t1=tuple(input(f"Enter a tuple:").split())

even = t1[0::2]
odd = t1[1::2]

print("even =", even)
print("odd =", odd)
```

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

In [5]:
```python
rows = int(input("Enter number of rows: "))

T = []

for i in range(rows):
    t1 = tuple(map(int, input(f"Enter tuple {i+1}: ").split()))
    T.append(t1)

T = tuple(T)

cols = len(T[0])

for j in range(cols):
    s = 0
    for i in range(rows):
        s += T[i][j]
    print(s, end=" ")
```

```
12 15 18
```

In [ ]: